

BCPD Python Developer

Practice-exam questions and answers

133 questions

Correct options are marked in **green**.

These are the practice-exam questions. Four keys disagreed with the interpreter when the code was run and are corrected here, each one saying what changed. There are no GUI Programming questions in this set, so it does not cover that part of the exam.

Practice-exam set (133)

Domain 1 — Introduction and Setup (37)

Introduction and Setup

Which of the following is not a version of Python?

- A. Python 2
- B. Python 3
- C. Python X**
- D. Python 4

Answer: C

From the tutor's practice bank (item 1, tagged BCPD). No explanation is published with it; the answer shown is the bank's own key.

Introduction and Setup

What will the value of the i variable be when the following loop finishes its execution?

```
for i in range(10):  
    pass
```

- A. 10
- B. the variable becomes unavailable
- C. 11
- D. 9**

Answer: D

From the tutor's practice bank (item 3, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Introduction and Setup

The following expression $1+-2$ is:

- A. equal to 1
- B. invalid
- C. equal to 2
- D. equal to -1**

Answer: D

From the tutor's practice bank (item 4, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Introduction and Setup

A compiler is a program designed to

- A. rearrange the source code to make it clearer
- B. check the source code in order to see if it's correct**
- C. execute the source code
- D. translate the source code into machine code**

Answer: B, D

From the tutor's practice bank (item 5, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Introduction and Setup

What is the output of the following piece of code?

```
a='ant'
b='bat'
c='camel'
print(a,b,c,sep='')
```

- A. ant' bat' camel
- B. ant'bat'camel
- C. antbatcamel**
- D. SyntaxError

Answer: C

From the tutor's practice bank (item 6, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Introduction and Setup

What is the expected output of the following snippet?

```
i=5
while i>0:
    i=i//2
    if i%2==0:
        break
    else:
        i+=1
print(i)
```

- A. the code is erroneous**
- B. 3
- C. 7
- D. 15

Answer: A

From the tutor's practice bank (item 7, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Introduction and Setup

How many lines does the following snippet output?

```
for i in range(1,3):
    print('**',end='')
else:
    print('**')
```

- A. three
- B. one
- C. two**
- D. four

Answer: C

From the tutor's practice bank (item 8, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Introduction and Setup

Which of the following literals reflect the value given as 34.23?

- A. 3.42E+01**
- B. 3.42E+01**
- C. 3.42E-03

D. 3.42E+05

Answer: A, B

From the tutor's practice bank (item 9, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Introduction and Setup

Select the valid fun() invocations:

```
def fun(a,b=0):  
    return a*b
```

- A. fun(b=1)
- B. fun(a=0)**
- C. fun(b=1,0)
- D. fun(1)**

Answer: B, D

From the tutor's practice bank (item 22, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Introduction and Setup

What is the expected output of the following code?

```
def f(n):  
    if n==1:  
        return '1'  
    return str(n)+f(n-1)  
print(f(2))
```

- A. 21**
- B. 12
- C. 3
- D. 1

Answer: A

From the tutor's practice bank (item 27, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Introduction and Setup

What is the expected behaviour of the following snippet?

```
def x():  
    return 2  
x=1+x()  
print(x)
```

- A. cause a runtime exception on line 02
- B. cause a runtime exception on line 01
- C. cause a runtime exception on line 03
- D. print 3**

Answer: D

From the tutor's practice bank (item 28, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Introduction and Setup

What is the expected behaviour of the following code?

```
def f(n):  
    for i in range(1,n+1):  
        yield i  
print(f(2))
```

- A. print 4321**

- B. print <generator object f at (some hex digits)>
- C. cause a runtime exception
- D. print 1234

Answer: B

From the tutor's practice bank (item 29, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Introduction and Setup

If you need a function that does nothing, what would you use instead of XXX?

```
| def idler(): XXX
```

- A. pass
- B. return
- C. exit
- D. None

Answer: A, B

From the tutor's practice bank (item 30, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Introduction and Setup

Which of the following words can be used as a variable name? (Select two valid names)

- A. for
- B. TRUE
- C. TRUE
- D. For

Answer: C, D

From the tutor's practice bank (item 41, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Introduction and Setup

Python strings can be 'glued' together using the operator:

- A. &
- B. *
- C. ",
- D. +

Answer: D

From the tutor's practice bank (item 42, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Introduction and Setup

A keyword

- A. can be used as an identifier
- B. is defined by Python's lexis
- C. is also known as a reserved word
- D. cannot be used in the user's code

Answer: B, C

From the tutor's practice bank (item 43, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Introduction and Setup

How many stars (*) does the snippet print?

```
| S='*****'  
| print(s)
```

- A. the code is erroneous
- B. five

- C. four
- D. None (11 stars)

Answer: A

From the tutor's practice bank (item 44, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Introduction and Setup

Assuming that V holds integer 2, which operator should be used instead of OPER to make V OPER 1 equal to 1?

- A. «<
- B. »>
- C. »
- D. «

Answer: C

From the tutor's practice bank (item 46, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Introduction and Setup

How many stars (*) does the following snippet print?

```
i=3
while i>0:
    i-=1
    print('*')
else:
    print('*')
```

- A. the code is erroneous
- B. five
- C. three
- D. four

Answer: D

From the tutor's practice bank (item 47, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Introduction and Setup

UNICODE is:

- A. the name of an operating system
- B. a standard for encoding and handling texts
- C. the name of a programming language
- D. the name of a text processor

Answer: B

From the tutor's practice bank (item 48, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Introduction and Setup

Which of the equations are True?

- A. chr(ord(x)) == x
- B. ord(ord(x)) == x
- C. chr(chr(x)) == x
- D. ord(chr(x)) == x

Answer: A, D

From the tutor's practice bank (item 53, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Introduction and Setup

A two-parameter lambda function raising its first parameter to the power of the second parameter should be declared as:

- A. `lambda (x,y) = x**y`
- B. `lambda (x,y): x**y`
- C. `def lambda (x,y): return x**y`
- D. `lambda x, y: x**y`**

Answer: D

From the tutor's practice bank (item 60, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Introduction and Setup

What is the expected output of the following code?

```
def f(n):  
    if n==1:  
        return 1  
    return n+f(n-1)  
print(f(2))
```

- A. 21
- B. 12
- C. 3**
- D. none

Answer: C

From the tutor's practice bank (item 61, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Introduction and Setup

Which function or operator should you use to obtain the answer True or False to the question: 'Do two variables refer to the same object?'

- A. The = operator
- B. The `isinstanceOf` function
- C. The `id()` function
- D. The `is` operator**

Answer: D

From the tutor's practice bank (item 82, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Introduction and Setup

Select the true statements related to PEP 8 programming recommendations.

- A. You should use the `not ... is` operator rather than the `is not` operator
- B. You should make object type comparisons using the `isinstance()` method**
- C. You should write code in a way that favours the CPython implementation over PyPy, Cython, and Jython
- D. You should not write string literals that rely on significant trailing whitespace**

Answer: B, D

From the tutor's practice bank (item 85, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Introduction and Setup

Which of the following statements are true? (Choose two.)

- A. ASCII stands for Information Interchange**
- B. a code point is a number assigned to a given character**
- C. ASCII is synonymous with UTF-8
- D. `\e` is an escape sequence used to mark the end of lines

Answer: A, B

From the tutor's practice bank (item 88, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Introduction and Setup

What is the expected output of the following code?

```
myTu=('a','b','c')
m=tuple(map(lambda x: chr(ord(x)+1), myTu))
print(m[-2])
```

- A. a
- B. c**
- C. an exception is raised
- D. b

Answer: B

Corrected. The bank keys b. Running it: ('a','b','c') maps through chr(ord(x)+1) to ('b','c','d'), so m[-2] is c. Option B.

Introduction and Setup

Assuming that the following code has been executed successfully, which of the expressions evaluate to True? (Choose two.)

```
def f(x,y):
    nom,denom=x,y
    def g():
        return nom/denom
    return g
a=f(1,2)
b=f(3,4)
```

- A. b()==4
- B. a!=b**
- C. a is not None**
- D. a()==4

Answer: B, C

From the tutor's practice bank (item 109, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Introduction and Setup

def foo(x,y): return y(x)+(x+1) print(foo(1, lambda x: x*x))

```
def foo(x,y):
    return y(x)+(x+1)
print(foo(1, lambda x: x*x))
```

- A. 3**
- B. 5
- C. 4
- D. an exception is raised

Answer: A

From the tutor's practice bank (item 110, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Introduction and Setup

Which of the following lambda definitions are correct? (Choose two.)

- A. lambda x,y: (x,y)**
- B. lambda x,y: return x/y - x%y
- C. lambda x,y: x/y - x%y**
- D. lambda x,y = x/y - x%y

Answer: A, C

From the tutor's practice bank (item 111, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Introduction and Setup

`x = 3 % 1 y = 1 if x > 0 else 0 print(y)`

```
x = 3 % 1
y = 1 if x > 0 else 0
print(y)
```

- A. the code is erroneous and it will not execute
- B. it outputs 1
- C. it outputs -1
- D. it outputs 0**

Answer: D

From the tutor's practice bank (item 113, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Introduction and Setup

`x = 8 ** (1/3) y = 2. if x < 2.3 else 3. print(y)`

```
x = 8 ** (1/3)
y = 2. if x < 2.3 else 3.
print(y)
```

- A. the code is erroneous and it will not execute
- B. it outputs 2.0**
- C. it outputs 2.5
- D. it outputs 3.0

Answer: B

From the tutor's practice bank (item 117, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Introduction and Setup

Which of the following statements are true? (Choose two.)

- A. an escape sequence can be recognised by the / sign put in front of it
- B. ASCII stands for Internal Information
- C. ASCII is a subset of UNICODE**
- D. a code point is a number assigned to a given character**

Answer: C, D

From the tutor's practice bank (item 121, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Introduction and Setup

`string='123' dummy=0 for character in reversed(string): dummy += int(character) print(dummy)`

```
string='123'
dummy=0
for character in reversed(string):
    dummy += int(character)
print(dummy)
```

- A. it outputs 321
- B. it outputs 123
- C. it outputs 6**
- D. it raises an exception

Answer: C

From the tutor's practice bank (item 122, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Introduction and Setup

Which of the following expressions evaluate to True? (Choose two.)

- A. `ord('2') - ord('2') == ord('0')`
- B. `len('6E6E6E6E') > 0`
- C. `chr(ord('a')+1) == 'B'`
- D. `len('') == 1`

Answer: A, B

From the tutor's practice bank (item 124, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Introduction and Setup

`def foo(x,y): return y(x)+y(x+1) print(foo(1, lambda x: x*x))`

```
def foo(x,y):
    return y(x)+y(x+1)
print(foo(1, lambda x: x*x))
```

- A. 4
- B. 3
- C. an exception is raised
- D. 5

Answer: D

From the tutor's practice bank (item 133, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Introduction and Setup

Which of the following expressions evaluate to True? (Choose two.)

- A. `ord('2') - ord('2') == ord('0')`
- B. `chr(ord('A')+1) == 'B'`
- C. `len('') == 1`
- D. `len('0') == ('E'*'E')`

Answer: A, B

From the tutor's practice bank (item 145, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Domain 2 — Data Structures (20)

Data Structures

Assuming that the following snippet has been successfully executed, which of the equations are True?

```
a=[1]
b=a
a[0]=0
```

- A. `len(a)==len(b)`
- B. `b[0]+1==a[0]`
- C. `a[0]==b[0]`
- D. `a[0]+1==b[0]`

Answer: A, C

From the tutor's practice bank (item 11, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Data Structures

Assuming that the following snippet has been successfully executed, which of the equations are False?

```
a=[0]
b=a[: ]
a[0]=1
```

- A. `len(a)==len(b)`
- B. `a[0]-1==b[0]`
- C. `a[0]==b[0]`
- D. `b[0]-1==a[0]`

Answer: C, D

From the tutor's practice bank (item 12, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Data Structures

Which of the following statements are false?

- A. Python strings are actually lists
- B. Python strings can be concatenated
- C. Python strings can be sliced like lists
- D. Python strings are mutable

Answer: A, D

From the tutor's practice bank (item 13, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Data Structures

Which of the following sentences are true?

- A. Lists may not be stored inside tuples
- B. Tuples may be stored inside lists
- C. Tuples may not be stored inside tuples
- D. Lists may be stored inside lists

Answer: B, D

From the tutor's practice bank (item 14, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Data Structures

Assuming that String is six or more letters long, the following slice `string[1:-2]` is shorter than the original string by:

- A. four chars
- B. three chars
- C. one char
- D. two chars

Answer: B

From the tutor's practice bank (item 15, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Data Structures

What is the expected output of the following snippet?

```
lst=[1,2,3,4]
lst=lst[-3:-2]
lst=lst[-1]
print(lst)
```

- A. 1
- B. 4
- C. 2
- D. 3

Answer: C

From the tutor's practice bank (item 16, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Data Structures

How many elements will the list2 list contain after execution of the following snippet?

```
list1=[False for i in range(1,10)]  
list2=list1[-1:1:-1]
```

- A. zero
- B. five
- C. seven**
- D. three

Answer: C

From the tutor's practice bank (item 18, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Data Structures

What would you use instead of XXX if you want to check whether a certain key exists in a dictionary called dict?

- A. 'key' in dict**
- B. dict['key'] != None
- C. dict.exists('key')
- D. 'key' in dict.keys()**

Answer: A, D

From the tutor's practice bank (item 19, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Data Structures

You need data which can act as a simple telephone directory. You can obtain it with the following clauses (Select two relevant variants; assume no other items have been created)

- A. dir={'Mom':5551234567,'Dad':5557654321}**
- B. dir={'Mom':5551234567,'Dad':5557654321}**
- C. dir={'Mom':5551234567,'Dad':5557654321}
- D. dir={'Mom':5551234567,'Dad':5557654321}

Answer: A, B

From the tutor's practice bank (item 20, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Data Structures

What is the expected output of the following code?

```
str='abcdef'  
def fun(s):  
    del s[2]  
    return s  
print(fun(str))
```

- A. abcef
- B. The program will cause a runtime exception/error**
- C. acdef
- D. abdef

Answer: B

From the tutor's practice bank (item 26, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Data Structures

Which of the listed actions can be applied to the following tuple?

```
tup=()
```

- A. tup[0]
- B. tup.append(0)**

- C. `tup[0]`
- D. `del tup`

Answer: C, D

From the tutor's practice bank (item 50, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Data Structures

Executing the snippet `dct={'pi':3.14}; dct['pi']=3.1415` will cause the `dct`:

- A. to hold two keys named 'pi' linked to 3.14 and 3.1415 respectively
- B. to hold two key named 'pi' linked to 3.14 and 3.1415
- C. to hold one key named 'pi' linked to 3.1415
- D. to hold two keys named 'pi' linked to 3.1415

Answer: C

From the tutor's practice bank (item 51, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Data Structures

How many elements will the `list1` list contain after execution of the following snippet?

```
list1 = 'don't think twice, do it!'.split(',')
```

- A. two
- B. zero
- C. one
- D. three

Answer: A

From the tutor's practice bank (item 52, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Data Structures

If you want to transform a string into a list of words, what invocation would you use? (Expected output: The, Catcher, in, the Rye,)

```
S='The Catcher in the Rye'
L=# put proper invocation
```

- A. `s.split()`
- B. `split(s,')`
- C. `s.split('')`
- D. `split(s)`

Answer: A

From the tutor's practice bank (item 54, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Data Structures

Assuming that `lst` is a four-element list, is there any difference between these two statements? `del lst` `del lst[:]`

- A. yes, the first line empties the list, the second deletes the list as a whole
- B. yes, the first line deletes the list as a whole, the second just empties the list
- C. no, there is no difference
- D. yes, the first line deletes the list as a whole, the second removes all elements except the first

Answer: B

From the tutor's practice bank (item 55, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Data Structures

Which of the following expressions evaluate to True? (Choose two.)

- A. `str(1-1) in '123456789'`
- B. `'dcb' not in 'abcde'[::-1]`
- C. `'phd' in 'alpha'`
- D. `'True' not in 'False'`

Answer: A, D

From the tutor's practice bank (item 89, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Data Structures

What is the expected behaviour of the following code?

```
the_list='alpha;beta;gamma'.split(';')
the_string='.'.join(the_list)
print(the_string.isalpha())
```

- A. it raises an exception
- B. it outputs True
- C. it outputs False
- D. it outputs nothing

Answer: C

From the tutor's practice bank (item 90, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Data Structures

Assuming that the snippet below has been executed successfully, which of the following expressions evaluate to True? (Choose two.)

```
string='python'[:2]
string=string[-1]+string[-2]
```

- A. `string[0]=='o'`
- B. `string is None`
- C. `len(string)==3`
- D. `string[0]==string[-1]`

Answer: A

From the tutor's practice bank (item 92, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Data Structures

Which of the following expressions evaluate to True? (Choose two.)

- A. `'in' in 'Thames'`
- B. `'in' in 'in'`
- C. `'in not' in 'not'`
- D. `'t'.upper() in 'Thames'`

Answer: B, D

From the tutor's practice bank (item 136, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Data Structures

Which of the following invocations are valid? (Choose two.)

- A. `sort('python')`
- B. `'python'.find('e')`
- C. `'python'.index('eth')`
- D. `'python'.sorted()`

Answer: B, C

From the tutor's practice bank (item 138, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Domain 3 — Object-Oriented Programming (32)

Object-Oriented Programming

Is it possible to safely check if a class/object has a certain attribute?

- A. yes, by using the hasstart attribute
- B. yes, by using the hasstart() method
- C. yes, by using the hasattr() function**
- D. no, it is not possible

Answer: C

From the tutor's practice bank (item 31, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Object-Oriented Programming

The first parameter of each method:

- A. holds a reference to the currently processed object**
- B. is always set to None
- C. is set to a unique random value
- D. is set by the first argument's value

Answer: A

From the tutor's practice bank (item 32, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Object-Oriented Programming

The simplest possible class definition in Python can be expressed as:

- A. class X:
- B. class X: pass**
- C. class X: return
- D. class X: {}

Answer: B

From the tutor's practice bank (item 33, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Object-Oriented Programming

A variable stored separately in every object is called:

- A. there are no such variables, all variables are shared among objects
- B. a class variable
- C. an object variable
- D. an instance variable**

Answer: D

From the tutor's practice bank (item 35, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Object-Oriented Programming

The following class hierarchy is given. What is the expected output of the code?

```
class A:
    def a(self):
        print('A',end=' ')
    def b(self):
        self.a()
...
B().do()
C().do()
```

- A. BB
- B. CC
- C. AA

D. BC

Answer: D

From the tutor's practice bank (item 39, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Object-Oriented Programming

Scenario: You are developing a game in Python and want to represent different characters as objects. Which OOP concept will you use?

- A. Inheritance
- B. Encapsulation
- C. Polymorphism**
- D. Abstraction

Answer: C

From the tutor's practice bank (item 48, tagged BCPD). No explanation is published with it; the answer shown is the bank's own key.

Object-Oriented Programming

If any of a class's components has a name that starts with two underscores (`__`), then:

- A. the class component's name will be mangled**
- B. the class component has to be an instance variable
- C. the class component has to be a class variable
- D. the class component has to be a method

Answer: A

From the tutor's practice bank (item 66, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Object-Oriented Programming

A function called `issubclass(c1,c2)` is able to check if:

- A. c1 and c2 are both subclasses of the same superclass
- B. c2 is a subclass of c1
- C. c1 is a subclass of c2**
- D. c1 and c2 are not subclasses of the same superclass

Answer: C

From the tutor's practice bank (item 68, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Object-Oriented Programming

A class constructor

- A. can return a value
- B. cannot be invoked directly from inside the class**
- C. can be invoked directly from any of the subclasses**
- D. can be invoked directly from any of the superclasses

Answer: B, C

From the tutor's practice bank (item 69, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Object-Oriented Programming

The following class definition is given. We want the `show()` method to invoke the `get()` method, and then output the value the `get()` method returns. Which invocation should be used instead of XXX?

```
class Class:
    def __init__(self, val):
        self.val = val
    def get(self):
        return self.val
    def show(self):
        XXX
```

- A. `print(get(self))`
- B. `print(self.get())`**
- C. `print(get())`
- D. `print(self.get(val))`

Answer: B

From the tutor's practice bank (item 70, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Object-Oriented Programming

What is the expected output of the following snippet?

```
class X: pass
class Y(X): pass
class Z(Y): pass
X=Z()
Z=Z()
print(isinstance(X,Z), isinstance(Z,X))
```

- A. True False
- B. True True
- C. False False
- D. False True**

Answer: D

From the tutor's practice bank (item 73, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Object-Oriented Programming

Which sentence about the `@property` decorator is false?

- A. The `@property` decorator should be defined after the method that is responsible for setting an encapsulated attribute.**
- B. The `@property` decorator designates a method which is responsible for returning an attribute value.
- C. The `@property` decorator marks the method whose name will be used as the name of the instance attribute.
- D. The `@property` decorator should be defined before the methods that are responsible for setting and deleting an encapsulated attribute.

Answer: A

From the tutor's practice bank (item 83, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Object-Oriented Programming

Select the true statement about the `__name__` attribute.

- A. `__name__` is a special attribute, inherent for both classes and instances, and it contains information about the class to which a class instance belongs.
- B. `__name__` is a special attribute, inherent for both classes and instances, and it contains a dictionary of object attributes.
- C. `__name__` is a special attribute, inherent for classes and it contains information about the class to which a class instance belongs.
- D. `__name__` is a special attribute, inherent for classes, and it contains the name of a class.**

Answer: D

From the tutor's practice bank (item 84, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Object-Oriented Programming

What is the expected behaviour of the following code? (missing colon after `Sub_B(Super)`)

- A. it outputs 0
- B. it outputs 1
- C. it raises an exception**
- D. it outputs 2

Answer: C

From the tutor's practice bank (item 93, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Object-Oriented Programming

Assuming that the following inheritance set is in force, which of the following classes are declared properly? (Choose two.)

```
class A: pass
class B(A): pass
class C(A): pass
class D(B): pass
```

- A. class Class_4(D,A): pass
- B. class Class_3(A,C): pass
- C. class Class_2(B,D): pass**
- D. class Class_1(C,D): pass

Answer: A, C

From the tutor's practice bank (item 94, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Object-Oriented Programming

```
class Upper: def method(self): return 'upper'
class Lower(Upper): def method(self): return 'lower'
Object=Upper()
print(isinstance(Object,Upper), end=' ')
print(Object.method())
```

```
class Upper:
    def method(self):
        return 'upper'
class Lower(Upper):
    def method(self):
        return 'lower'
Object=Upper()
print(isinstance(Object,Lower), end=' ')
print(Object.method())
```

- A. True upper
- B. True lower
- C. False upper**
- D. False lower

Answer: C

From the tutor's practice bank (item 95, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Object-Oriented Programming

What is the expected behaviour of the following code? (o1.foo missing parentheses)

- A. it outputs 1
- B. it outputs 3
- C. it outputs 6
- D. it raises an exception**

Answer: D

From the tutor's practice bank (item 97, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Object-Oriented Programming

```
class Class: Variable=0
def __init__(self):
    self.value=0
object_1=Class()
object_1.Variable+=1
object_2=Class()
object_2.value+=1
print(object_2.Variable+object_1.value)
```

```
class Class:
    Variable=0
    def __init__(self):
        self.value=0
object_1=Class()
```

```
object_1.Variable+=1
object_2=Class()
object_2.value+=1
print(object_2.Variable+object_1.value)
```

- A. it outputs 0
- B. it raises an exception
- C. it outputs 1
- D. it outputs 2

Answer: A

From the tutor's practice bank (item 98, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Object-Oriented Programming

What is true about Object-Oriented Programming in Python? (Choose two.)

- A. each object of the same class can have a different set of methods
- B. a subclass is usually more specialised than its superclass
- C. if a real-life object can be described with a set of adjectives, they may reflect a Python object method
- D. the same class can be used many times to build a number of objects

Answer: B, D

From the tutor's practice bank (item 99, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Object-Oriented Programming

What is true about Python class constructors? (Choose two.)

- A. there can be more than one constructor in a Python class
- B. the constructor must return a value other than None
- C. the constructor is a method named `__init__`
- D. the constructor must have at least one parameter

Answer: C, D

From the tutor's practice bank (item 100, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Object-Oriented Programming

Assuming that the following piece of code has been executed successfully, which of the expressions evaluate to True? (Choose two.)

```
class A:
    VarA=1
    def __init__(self):
        self.prop_a=1
class B(A):
    VarA=2
    def __init__(self):
        super().__init__()
        self.prop_b=2
obj_a=A()
obj_aa=A()
obj_b=B()
obj_bb=obj_b
```

- A. `isinstance(obj_b,A)`
- B. `A.VarA==1`
- C. `obj_a is obj_aa`
- D. `B.VarA==1`

Answer: A, B

From the tutor's practice bank (item 101, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Object-Oriented Programming

class Class: Var=data=1 def __init__(self, value): self.prop=value Object=Class(2) Which are true? (Choose two.)

```
class Class:
    Var=data=1
    def __init__(self, value):
        self.prop=value
Object=Class(2)
Which are true? (Choose two.)
```

- A. len(Class.__dict__)==1
- B. 'data' in Class.__dict__**
- C. 'var' in Class.__dict__
- D. 'data' in Object.__dict__

Answer: B

From the tutor's practice bank (item 102, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Object-Oriented Programming

A property that stores information about a given class's super-classes is named:

- A. __upper__
- B. __super__
- C. __ancestors__
- D. __bases__**

Answer: D

From the tutor's practice bank (item 103, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Object-Oriented Programming

class Class: Var=0 def _foo(self): Class.Var+=1 return Class.Var o=Class() o._Class_foo() print(o._Class_foo())

```
class Class:
    Var=0
    def _foo(self):
        Class.Var+=1
        return Class.Var
o=Class()
o._Class_foo()
print(o._Class_foo())
```

- A. it raises an exception**
- B. it outputs 3
- C. it outputs 1
- D. it outputs 6

Answer: A

From the tutor's practice bank (item 116, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Object-Oriented Programming

The __bases__ property contains:

- A. base class location (addr)
- B. base class objects (class)**
- C. base class names (str)
- D. base class ids (int)

Answer: B

From the tutor's practice bank (item 119, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Object-Oriented Programming

```
class Super:
    def make(self):
        pass
    def doit(self):
        return self.make()
class Sub_A(Super):
    def make(self):
        return 1
class Sub_B(Super):
    pass
a=Sub_A()
b=Sub_B()
print(a.doit()+b.doit())
```

```
class Super:
    def make(self):
        pass
    def doit(self):
        return self.make()
class Sub_A(Super):
    def make(self):
        return 1
class Sub_B(Super):
    pass
a=Sub_A()
b=Sub_B()
print(a.doit()+b.doit())
```

- A. It outputs 0
- B. It raises an exception**
- C. It outputs 1
- D. It outputs 2

Answer: B

From the tutor's practice bank (item 120, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Object-Oriented Programming

Assuming that the code below has been executed successfully, which of the expressions evaluate to True? (Choose two.)

```
class Class:
    var=1
    def __init__(self, value):
        self.prop=value
Object=Class(2)
```

- A. 'var' in Class.__dict__**
- B. 'var' in Object.__dict__
- C. len(Object.__dict__) == 1**
- D. 'prop' in Class.__dict__

Answer: A, C

From the tutor's practice bank (item 126, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Object-Oriented Programming

What is true about Python class constructors? (Choose two.)

- A. there can be only one constructor in a Python class
- B. the constructor cannot be invoked directly under any circumstances**
- C. the constructor cannot return a result other than None
- D. the constructor's first parameter must always be named self**

Answer: B, D

From the tutor's practice bank (item 132, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Object-Oriented Programming

Assuming that the following inheritance set is in force, which of the following classes are declared properly? (Choose two.)

```
class A:
    pass
class B(A):
    pass
class C(A):
    pass
class D(B,C):
    pass
```

- A. class Class_3(A,C): pass
- B. class Class_4(C,B): pass
- C. class Class_1(D): pass
- D. class Class_2(A,B): pass

Answer: A, C

From the tutor's practice bank (item 134, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Object-Oriented Programming

```
class Class: Variable=0 def __init__(self): self.value=0 object_1=Class() Class.Variable+=1 object_2=Class() object_2.value+=1
print(object_2.Variable+object_1.value)
```

```
class Class:
    Variable=0
    def __init__(self):
        self.value=0
object_1=Class()
Class.Variable+=1
object_2=Class()
object_2.value+=1
print(object_2.Variable+object_1.value)
```

- A. It outputs 2
- B. It raises an exception
- C. It outputs 1
- D. It outputs 0

Answer: C

From the tutor's practice bank (item 137, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Object-Oriented Programming

```
class Upper: def __init__(self): self.property='upper' class Lower(Upper): def __init__(self): super().__init__() Object=Lower()
print(isinstance(Object,Lower), end='') print(Object.property)
```

```
class Upper:
    def __init__(self):
        self.property='upper'
class Lower(Upper):
    def __init__(self):
        super().__init__()
Object=Lower()
print(isinstance(Object,Lower), end='')
print(Object.property)
```

- A. True lower
- B. True upper
- C. False upper
- D. False lower

Answer: B

From the tutor's practice bank (item 139, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Object-Oriented Programming

```
class ClassA: var=1 def __init__(self,prop): prop1=prop2=prop class ClassB(ClassA): def __init__(self,prop): prop3=prop**2
super().__init__(prop) def __str__(self): return 'Object' Object=ClassA(2)
```

```
class ClassA:
    var=1
    def __init__(self,prop):
        prop1=prop2=prop
class ClassB(ClassA):
    def __init__(self,prop):
```

```

        prop3=prop**2
        super().__init__(prop)
    def __str__(self):
        return 'Object'
Object=ClassA(2)

```

- A. `ClassA.__module__ == '__main__'`
- B. `__name__ == '__main__'`
- C. `str(Object) == 'Object'`
- D. `len(ClassB.__bases__) == 2`

Answer: A, C

From the tutor's practice bank (item 141, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Domain 4 — Modules and Libraries (17)

Modules and Libraries

Can a module run like regular code?

- A. yes, and it can differentiate its behaviour between the regular launch and import
- B. it depends on the Python version
- C. yes, but it cannot differentiate its behaviour between the regular launch and import
- D. no, it is not possible; a module can be imported, not run

Answer: A

From the tutor's practice bank (item 21, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Modules and Libraries

A file name like `services.cpython-36.pyc` says that:

- A. the interpreter used to generate the file is version 3.6
- B. it has been produced by CPython
- C. it is the 36th version of the file
- D. the file comes from the `services.py` source file

Answer: A, B, D

From the tutor's practice bank (item 23, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Modules and Libraries

What can you do if you don't like a long package path like `import alpha.beta.gamma.delta.epsilon.zeta`?

- A. you can make an alias for the name using the `alias` keyword
- B. nothing, you need to come to terms with it
- C. you can shorten it to `alpha.zeta` and Python will find the proper connection
- D. you can make an alias for the name using the `as` keyword

Answer: D

From the tutor's practice bank (item 25, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Modules and Libraries

What should you put instead of XXX to print out the module name?

```

if __name__ != 'XXX':
    print(__name__)

```

- A. `main`
- B. `_main`
- C. `__main__`
- D. `_main_`

Answer: C

From the tutor's practice bank (item 56, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Modules and Libraries

Files with the suffix .py contain:

- A. Python source code
- B. backups
- C. temporary data
- D. semi-compiled Python code

Answer: A

From the tutor's practice bank (item 57, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Modules and Libraries

Package source directories/folders can be:

- A. converted into the so-called pyck format
- B. packed as a ZIP file and distributed as one file
- C. rebuilt to a flat form and distributed as one directory/folder
- D. removed as Python compiles them into an internal portable format

Answer: B

From the tutor's practice bank (item 58, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Modules and Libraries

What can you deduce from the line below?

```
x = a.b.c.f()
```

- A. import a.b.c should be placed before that line
- B. f() is located in subpackage c of subpackage b of package a
- C. the line is incorrect
- D. the function being invoked is called a.b.c.f()

Answer: A, B, D

From the tutor's practice bank (item 59, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Modules and Libraries

Assuming that the code below has been executed successfully, which of the following expressions will always evaluate to True? (Choose two.)

```
import random
random.seed(1)
v1=random.random()
random.seed(1)
v2=random.random()
```

- A. v1 == v2
- B. v1 > v2
- C. len(random.sample([1,2,3],2)) > 2
- D. random.choice([1,2,3]) >= 1

Answer: A, D

From the tutor's practice bank (item 74, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Modules and Libraries

Which one of the platform module functions should be used to determine the underlying platform name?

- A. platform.python_version()
- B. platform.processor()
- C. platform.platform()
- D. platform.uname()

Answer: C

From the tutor's practice bank (item 75, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Modules and Libraries

What is the expected output of the following code?

```
import sys
import math
b1=type(dir(math)[0]) is str
b2=type(sys.path[-1]) is str
print(b1 and b2)
```

- A. FALSE
- B. None
- C. TRUE
- D. 0

Answer: C

From the tutor's practice bank (item 76, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Modules and Libraries

With regards to the directory structure, select the proper forms of the directives in order to import module_a. (Choose two.)

- A. from pypack import module_a
- B. import module_a from pypack
- C. import module_a
- D. import pypack.module_a

Answer: A, D

From the tutor's practice bank (item 77, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Modules and Libraries

A Python module named pymod.py contains a function named python(). Which of the following snippets will let you invoke the function? (Choose two.)

- A. import pymod; pymod.python()
- B. from pymod import python; python()
- C. from pymod import *; pymod.python()
- D. import python from pymod; python()

Answer: A, B

From the tutor's practice bank (item 78, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Modules and Libraries

What is true about Python packages? (Choose two.)

- A. a package is a single file whose name ends with the .pa extension
- B. a package is a group of related modules
- C. the __name__ variable always contains the name of a package
- D. the .pyc extension is used to mark semi-compiled Python packages

Answer: B, C

From the tutor's practice bank (item 79, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Modules and Libraries

A Python module named `pymod.py` contains a variable named `pymod`. Which of the following snippets will let you access the variable? (Choose two.)

- A. `import pymod; pymod.pyvar = 1`
- B. `import pymod from pymod; pymod = 1`
- C. `from pymod import pymod; pymod()`
- D. `from pymod import pymod; pymod = 1`

Answer: C, D

From the tutor's practice bank (item 123, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Modules and Libraries

Assuming that the code below has been executed successfully, which of the following expressions will always evaluate to True? (Choose two.)

```
import random
v1=random.random()
v2=random.random()
```

- A. `v1 == v2`
- B. `v1 < 1`
- C. `random.choice([1,2,3]) > 0`
- D. `len(random.sample([1,2,3],1)) > 2`

Answer: B, C

From the tutor's practice bank (item 128, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Modules and Libraries

With regards to the directory structure, select the proper forms of the directives in order to import `module_b`. (Choose two.)

- A. `from pyback.upper import module_b`
- B. `import pyback.upper.module_b`
- C. `import upper.module_b`
- D. `import module_b`

Answer: B

From the tutor's practice bank (item 129, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Modules and Libraries

`import sys` `b1=type(dir(sys))` is str `b2=type(sys.path[-1])` is str `print(b1 and b2)`

```
import sys
b1=type(dir(sys)) is str
b2=type(sys.path[-1]) is str
print(b1 and b2)
```

- A. FALSE
- B. 0
- C. None
- D. TRUE

Answer: A

The bank's key for this one reads FALSE — the option `*text*` rather than a letter, and almost certainly an Excel coercion of "False" into a boolean. It maps to option A. Running it confirms: `dir(sys)` returns a list, not a str, so `b1` is False and `b1 and b2` is False.

Domain 5 — Debugging and Error Handling (16)

Debugging and Error Handling

What is the expected output of the following snippet?

```
s='abc'
for i in len(s):
    s[i]=s[i].upper()
print(s)
```

- A. abc
- B. The code will cause a runtime exception**
- C. ABC
- D. 123

Answer: B

From the tutor's practice bank (item 17, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Debugging and Error Handling

What is the expected behaviour of the following snippet?

```
def a(l,l):
    return l[l]
print(a(0,[1]))
```

- A. cause a runtime exception
- B. print 1
- C. print 0, [1]
- D. SyntaxError**

Answer: D

From the tutor's practice bank (item 24, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Debugging and Error Handling

How to handle an exception and assign it to variable e?

- A. except Exception (e):
- B. except e = Exception:
- C. except Exception as e:**
- D. such an action is not possible in Python

Answer: C

From the tutor's practice bank (item 34, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Debugging and Error Handling

What is the expected output of the following code?

```
lst=[x for x in range(5)]
lst=list(filter(lambda x: x%2==0, lst))
print(len(lst))
```

- A. 2**
- B. The code will cause a runtime exception
- C. 1
- D. SyntaxError

Answer: A

From the tutor's practice bank (item 64, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Debugging and Error Handling

What is the expected behaviour of the following code?

```
def unclear(x):  
    if x%2==1:  
        return 0  
print(unclear(1)+unclear(2))
```

- A. print 0
- B. cause a runtime exception**
- C. prints 3
- D. print an empty line

Answer: B

From the tutor's practice bank (item 65, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Debugging and Error Handling

If you need to serve two different exceptions called Ex1 and Ex2 in one except branch, you can write:

- A. except Ex1 Ex2:
- B. except (Ex1, Ex2):**
- C. except Ex1, Ex2:
- D. except Ex1+Ex2:

Answer: B

From the tutor's practice bank (item 67, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Debugging and Error Handling

What is the expected behaviour of the following code? (try/except missing colon)

- A. it outputs 3
- B. it outputs 1
- C. the code is erroneous and it will not execute**
- D. it outputs 2

Answer: C

From the tutor's practice bank (item 80, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Debugging and Error Handling

What is the expected behaviour of the following code?

```
string=str(1/3)  
dummy=''  
for character in strong:  
    dummy=character+dummy  
print(dummy[-1])
```

- A. it raises an exception**
- B. it outputs 0
- C. it outputs 3
- D. it outputs None

Answer: A

From the tutor's practice bank (item 87, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Debugging and Error Handling

`my_list=[i for i in range(5)] m=[my_list[i] for i in range(4,0,-1)] if my_list[i]%2!=0] print(m)`

```
my_list=[i for i in range(5)]  
m=[my_list[i] for i in range(4,0,-1)] if my_list[i]%2!=0]  
print(m)
```

- A. it outputs [1,3]
- B. the code is erroneous and it will not execute**
- C. it outputs [3,1]
- D. it outputs [4,2,0]

Answer: B

From the tutor's practice bank (item 108, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Debugging and Error Handling

Which of the following snippets will execute without raising any unhandled exceptions? (Choose two.)

- A. try: print(float('1e1')) except (ValueError,NameError): print(float('1a1')) else: print(float('101'))**
- B. try: print(1/1) except: print(2/1) else: print(3/0)
- C. try: print(1/0) except ValueError: print(1/1) else: print(1/2)
- D. try: print(0/1) except: print(1/1) else: print(2/1)**

Answer: A, D

From the tutor's practice bank (item 114, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Debugging and Error Handling

`my_list=[i for i in range(5,0,-1)] m=[my_list[i] for i in range(5)] if my_list[i]%2==0] print(m)`

```
my_list=[i for i in range(5,0,-1)]
m=[my_list[i] for i in range(5)] if my_list[i]%2==0]
print(m)
```

- A. it outputs [4,2]
- B. it outputs [2,4]
- C. it outputs [0,1,2,3,4]
- D. the code is erroneous and it will not execute**

Answer: D

From the tutor's practice bank (item 118, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Debugging and Error Handling

`the_list='1,2 3'.split() the_string=''.join(the_list) print(the_string.insight())`

```
the_list='1,2 3'.split()
the_string=''.join(the_list)
print(the_string.insight())
```

- A. it raises an exception**
- B. it outputs nothing
- C. it outputs True
- D. it outputs False

Answer: A

From the tutor's practice bank (item 127, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Debugging and Error Handling

`s='2A' try: n=int(s) except ValueError: n=2 except ArithmeticError: n=1 except: n=0 print(n)`

```
s='2A'
try:
    n=int(s)
except ValueError:
    n=2
except ArithmeticError:
    n=1
except:
    n=0
print(n)
```

- A. it outputs 0
- B. the code is erroneous and it will not execute
- C. it outputs 1
- D. it outputs 2**

Answer: D

From the tutor's practice bank (item 135, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Debugging and Error Handling

Which of the following snippets will execute without raising any unhandled exceptions? (Choose two.)

- A. try: print(0/0) except: print(0/1) else: print(0/2)**
- B. try: print(int('0')) except NameError: print('0') else: print(int(''))
- C. import math try: print(math.sqrt(-1)) except: print(math.sqrt(0)) else: print(math.sqrt(1))**
- D. try: print(float('1e1')) except (NameError, SystemError): print(float('1a1')) else: print(float('1c1'))

Answer: A, C

From the tutor's practice bank (item 142, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Debugging and Error Handling

my_list=[1,2,3] try: my_list[3]=my_list[2] except BaseException as error: print(error)

```
my_list=[1,2,3]
try:
    my_list[3]=my_list[2]
except BaseException as error:
    print(error)
```

- A. it outputs error
- B. it outputs <class 'IndexError'>
- C. it outputs list assignment index out of range**
- D. the code is erroneous and it will not execute

Answer: C

From the tutor's practice bank (item 143, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Debugging and Error Handling

class E(Exception): def __init__(self,message): self.message=message def __str__(self): return 'it's nice to see you' try: print('I feel fine') raise Exception('what a pity') except E as e: print(e) else: print('the show must go on')

```
class E(Exception):
    def __init__(self,message):
        self.message=message
    def __str__(self):
        return 'it's nice to see you'
try:
    print('I feel fine')
    raise Exception('what a pity')
except E as e:
    print(e)
else:
    print('the show must go on')
```

- A. the string 'it's nice to see you' will be seen
- B. the string 'the show must go on' will be printed
- C. the string 'I feel fine' will be printed**
- D. nothing will be printed

Answer: C

From the tutor's practice bank (item 144, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

Domain 6 — File Handling (11)

File Handling

There is a stream named `s` open for writing. What option will you select to write a line to the stream?

- A. `s.write('Hello\n')`
- B. `write(s,'Hello')`
- C. `s.writeln('Hello')`
- D. `s.writeln('Hello')`

Answer: A

From the tutor's practice bank (item 36, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

File Handling

You are going to read just one character from a stream called `s`. Which statement would you use?

- A. `ch = read(s,1)`
- B. `ch = s.input(1)`
- C. `ch = input(s,1)`
- D. `ch = s.read(1)`

Answer: D

From the tutor's practice bank (item 37, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

File Handling

What can you deduce from the following statement?

```
| str = open('file.txt', 'rt')
```

- A. `str` is a string read in from the file named `file.txt`
- B. a newline character translation will be performed during the reads
- C. if `file.txt` does not exist, it will be created
- D. the opened file cannot be written with the use of the `str` variable

Answer: B, D

From the tutor's practice bank (item 38, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

File Handling

What does the `open()` function return?

- A. an integer value identifying an opened file
- B. an error code (0 means success)
- C. a stream object
- D. always `None`

Answer: C

From the tutor's practice bank (item 40, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

File Handling

If `S` is a stream open for reading, what do you expect from the following invocation?

```
| c = s.read()
```

- A. one line of the file will be read and stored in the string called `c`
- B. the whole file content will be read and stored in the string called `c`
- C. one character will be read and stored in the string called `c`
- D. one disk sector (512 bytes) will be read and stored in the string called `c`

Answer: B

From the tutor's practice bank (item 71, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

File Handling

You are going to read 16 bytes from a binary file into a bytearray called data. Which lines would you use?

- A. `data=bytearray(16); bf.readinto(data)`
- B. `data=binfile.read(bytearray(16))`
- C. `bf.readinto(data=bytearray(16))`
- D. `data=bytearray(binfile.read(16))`

Answer: A, D

From the tutor's practice bank (item 72, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

File Handling

Story: Sarah is writing a Python program that reads data from a file. Which of the following modes should she use to open the file for reading?

- A. `'w'`
- B. `'r'`
- C. `'a'`
- D. `'x'`

Answer: B

From the tutor's practice bank (item 93, tagged BCPD). No explanation is published with it; the answer shown is the bank's own key.

File Handling

try: `f=open('non_existing_file','w')` print(1,end=' ') `s=f.readline()` print(2,end=' ') except IOError as error: print(3,end=' ') else: `f.close()` print(4,end=' ')

```
try:
    f=open('non_existing_file','w')
    print(1,end=' ')
    s=f.readline()
    print(2,end=' ')
except IOError as error:
    print(3,end=' ')
else:
    f.close()
    print(4,end=' ')
```

- A. 124
- B. 1234
- C. 24
- D. 13

Answer: D

Corrected. The bank keys 124. Running it prints 1 3. Opening in 'w' mode succeeds and creates the file, so `print(1)` runs; then `readline()` on a write-only handle raises `io.UnsupportedOperation`, which subclasses `OSError` and so is caught by `except IOError`, printing 3. The `else` branch never runs because an exception was handled.

File Handling

try: `f=open('existing_text_file','w')` `d=f.readlines()` print(len(d)) `f.close()` except IOError: print(-1)

```
try:
    f=open('existing_text_file','w')
    d=f.readlines()
    print(len(d))
    f.close()
except IOError:
    print(-1)
```

- A. the length of the first line from the file
- B. -1
- C. the number of lines contained inside the file

D. the length of the last line from the file

Answer: B

The bank's key for this one reads 0 (none of the above), which is not one of the four options. Running it prints -1: 'w' mode truncates the file, `readlines()` on a write-only handle raises `io.UnsupportedOperation` (a subclass of `IOError`), the handler catches it and prints -1. Option B.

File Handling

Which of the following statements are true? (Choose two.)

- A. if invoking `open()` fails, an exception is raised
- B. `open()` requires a second argument
- C. `open()` is a function which returns an object that represents a physical file
- D. `instd`, `outstd`, `errstd` are the names of pre-opened streams

Answer: A, C

From the tutor's practice bank (item 112, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.

File Handling

try: f=open('non_zero_length_existing_text_file','rt') d=f.read(1) print(len(d)) f.close() except IOError: print(-1)

```
try:
    f=open('non_zero_length_existing_text_file','rt')
    d=f.read(1)
    print(len(d))
    f.close()
except IOError:
    print(-1)
```

- A. 1
- B. 0
- C. an error value corresponding to file not found
- D. 1

Answer: A

From the tutor's practice bank (item 125, tagged PCAP). No explanation is published with it; the answer shown is the bank's own key.